Problem 4

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Let F_n be the nth term in the Fibonacci sequence, $\{F_n\}$, defined as follows:

$$F_n = \begin{cases} 1 & \text{if } n = 1 \\ 1 & \text{if } n = 2 \\ F_{n-1} + F_{n-2} & \text{if } n \geq 3 \end{cases}$$

Then, let $\{O_n\}$ be the sequence of the Fibonacci sequence's *odd*-indexed terms and $\{E_n\}$ be the sequence of its *even*-indexed terms.

Let's observe the sums of these two sequences.

n	O_n	$\sum O_n$	E_n	$\sum E_n$
1	1	1	1	1
2	2	3	3	4
3	5	8	8	12
4	13	21	21	33
5	34	55	55	88
6	89	144	144	232
7	233	377	377	609
8	610	987	987	1596
9	1597	2584	2584	4180
10	4181	6765	6765	10945
11	10946	17711	17711	28656
12	28657	46368	46368	75024

Note that in the table's header I've abbreviated $\sum_{i=1}^{n} O_i$ to just $\sum O_n$. I've done the same for E_n as well.

From this table, you may see that the sum of one sequence is related to the terms of the other. Firstly, $\sum_{i=1}^{n} O_i = E_n$. Secondly, $\sum_{i=1}^{n} E_i = O_{n+1} - 1$. We will prove these identities in this document.

It would be more helpful to prove something about the Fibonacci sequence in general, so we should use $\{F_n\}$ in these proofs instead of $\{O_n\}$ and $\{E_n\}$. Fortunately, translating between these sequences isn't hard. For the odd-indexed terms, $O_j = F_{2j-1}$, and for the even-indexed terms, $E_j = F_{2j}$.

With that, let's dive into the proofs.

Proposition 1.

$$\forall m \in \mathbb{N}, \sum_{j=1}^{m} F_{2j-1} = F_{2m}$$

Proof. The proof is by induction on m.

BASE CASE: Consider the case for $m = 1$. Observe,

$$\begin{aligned}\sum_{j=1}^{1} F_{2j-1} &= F_{2(1)-1} \\ &= F_{2-1} \\ &= F_1 \\ &= 1\end{aligned}$$

and,

$$F_{2(1)} = F_2 = 1$$

Thus, the result holds for the base case $m = 1$.

INDUCTION: Suppose the result holds for some natural number k. That is,

$$\sum_{j=1}^{k} F_{2j-1} = F_{2k}$$

Now consider the case for $k + 1$. Observe,

$$\begin{aligned}\sum_{j=1}^{k+1} F_{2j-1} &= \sum_{j=1}^{k} (F_{2j-1}) + F_{2(k+1)-1} \\ &= F_{2k} + F_{2(k+1)-1} && \text{(by the inductive hypothesis)} \\ &= F_{2k} + F_{2k+1} \\ &= F_{2k+2} && \text{(by the definition of } F_n\text{)} \\ &= F_{2(k+1)}\end{aligned}$$

Notice that the substitution of F_{2k+2} for $F_{2k} + F_{2k+1}$ is valid because the index $2k + 2$ is always greater than 3. So F_{2k+2} sits within the $n \geq 3$ piece of F_n, where $F_n = F_{n-1} + F_{n-2}$.

The result, $F_{2(k+1)}$, has the form F_{2m} for $m = k + 1$. Thus, the result holds in the inductive case. Therefore, by the principle of mathematical induction, the result holds for any natural number m.

□

Proposition 2.

$$\forall m \in \mathbb{N}, \sum_{j=1}^{m} F_{2j} = F_{2m+1} - 1$$

Proof. The proof is by induction on m.

BASE CASE: Consider the case where $m = 1$. Observe,

$$\begin{aligned} \sum_{j=1}^{1} F_{2j} &= F_{2(1)} \\ &= F_2 \\ &= 1 \end{aligned}$$

and,

$$\begin{aligned} F_{2(1)+1} - 1 &= F_3 - 1 \\ &= 2 - 1 \\ &= 1 \end{aligned}$$

Therefore, the result holds for the base case $m = 1$.

INDUCTION: Suppose the result holds for some natural number k. That is,

$$\sum_{2j}^{k} = F_{2k+1} - 1$$

Consider the case for $k + 1$. Observe,

$$\begin{aligned} \sum_{j=1}^{k+1} F_{2j} &= \sum_{j=1}^{k} (F_{2j}) + F_{2(k+1)} \\ &= (F_{2k+1} - 1) + F_{2(k+1)} && \text{(by the inductive hypothesis)} \\ &= F_{2k+1} + F_{2k+2} - 1 \\ &= F_{2k+3} - 1 && \text{(by the definition of } F_n\text{)} \\ &= F_{2(k+1)+1} - 1 \end{aligned}$$

Once again, the substitution of F_{2k+3} for $F_{2k+1} + F_{2k+2}$ is valid because the index $2k + 3$ is always greater than 3.

The result $F_{2(k+1)+1} - 1$ has the form $F_{2m+1} - 1$ for $m = k + 1$. Thus, the result holds for the inductive case, and by the principle of mathematical induction the result holds for all $m \in \mathbb{N}$.

□

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